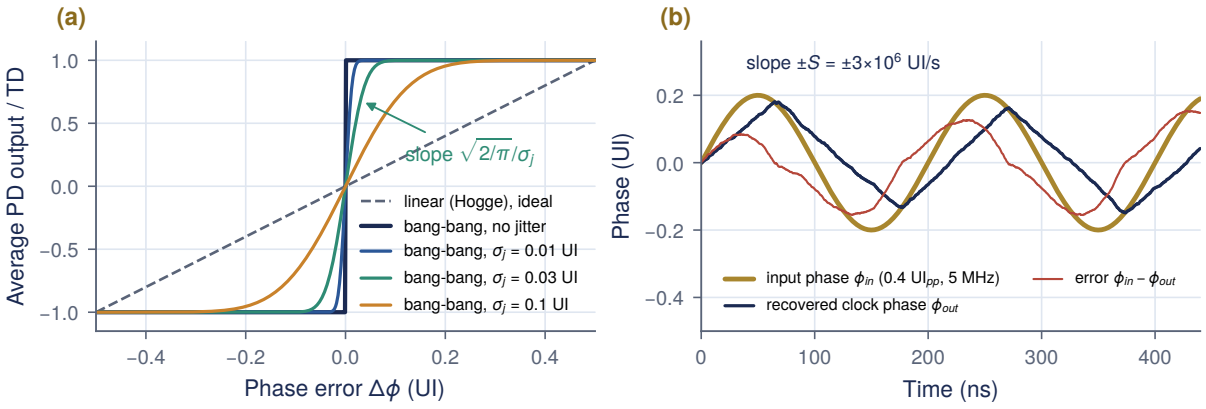
(a)
Average PD output / TD
1.0
0.5
0.0
−0.5
−1.0
−0.4
−0.2
0.0
0.2
0.4
Phase error Δϕ (UI)
slope √2/π/σj
linear (Hogge), ideal
bang-bang, no jitter
bang-bang, σj = 0.01 UI
bang-bang, σj = 0.03 UI
bang-bang, σj = 0.1 UI
(b)
slope ±S = ±3×10^6 UI/s
Phase (UI)
0.2
0.0
−0.2
−0.4
0
100
200
300
400
Time (ns)
input phase ϕin (0.4 UIpp, 5 MHz)
recovered clock phase ϕout
error ϕin − ϕout